

High FPS Video Frame Interpolation Challenge at NTIRE 2026

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Abstract

This paper presents a comprehensive review of the NTIRE 2026 High FPS Video Frame Interpolation Challenge, highlighting the proposed solutions and the final results. The objective of this challenge is to identify effective networks capable of producing clearer and visually compelling images in diverse and challenging conditions by learning representative visual cues for complex aggregations of motion types. A total of 79 participants registered for the first track and 44 for the second track of the competition, and 8 teams ultimately submitted valid entries. This paper thoroughly evaluates the state-of-the-art advances in high-FPS video frame interpolation, showcasing the significant progress in the field, while leveraging samples of the novel MIORe dataset that introduces challenging movement patterns.

1. Introduction

Video recordings are one of the most widespread media types across the internet. Thanks to the latest advancements in handheld devices' cameras, more and more people have access to professional quality recording means. That in turn encourages the use of mobile cameras in the everyday life, while it reduces the need of experience to choose the optimal parameters for a good capture. Although advanced, this automatization has offered regular users the opportunity to record videos of increasingly complexity while preserving the best quality possible. Nevertheless, this exposed the limitations of hardware components, as most consumer-

grade cameras cannot deal with either the sheer speed of subjects moving throughout the scene - resulting in motion blur, or with the frame stabilization, given the high degree of freedom one could manipulate the camera in.

Despite being a core component of videos, motion is predominantly understudied, since the existing state of knowledge revolves around uniform motion [24], isolated ego-motion [43], or environment-limited applications [7, 15]. Recently, some emerging works approached motion of higher intensity using high framerate cameras [10, 50], but the interest remains bound to crowded spaces and consumer-grade devices [46, 52].

Thus, most approaches focus on urban environments, as their intent is to model humans, vehicles, and other subjects as well as possible. The greatest interest lies within motion that violates the assumptions of correspondence, smoothness, or visibility. Preponderantly, the play between the layers of an urban scene, and the interactions between the subjects are observed from afar, under good conditions, and at a moderate speed. That is unfortunately not sufficient to capture the difficulty of non-rigid bodies, describe articulated motion, track large displacements, and solve the occlusion-disocclusion dynamics, as the importance of this sort of phenomena of critical relevance may be overseen in overcrowded videos, since the most attention would be paid to the overall scene quality, rather than to individual subjects displaying niche types of motion.

Furthermore, challenging environments often capture thin structures, fine geometry, object transparency, layered motion and specular & non-lambertian effects. Additionally, further increasing the difficulty are the rotational & out-of-plane motion (3D effects), motion blur, and the camera-induced complex motion.

Motion Patterns emerge from following the position and orientation in space of physical entities over time. Natu-

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Appendix A provides names and affiliations.

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rally, when considering a uni-dimensional point in space, there are only possibilities to translate it on its plane. However, when a second dimension is added, rotations become relevant as well. Ultimately, 3D objects have an increased complexity, and for a given entity, the most important types of movement are the translations (Ox, Oy, and Oz) and rotations (yaw, pitch, and roll). In addition, we need to also consider the behavior of the ego-camera, which is mainly influenced by its characteristic chaotic shake, especially for handheld devices [10]. Because of all the imperfections of the real world and the limitless possibilities of composing these base movement types, the need of intelligent methods for understanding our complex world becomes obvious.

The dataset we based our challenge on attempts to include several of these difficult components as either sole subjects or background presences, to offer participants the chance to experiment with different temporal superresolution scenarios.

The task of Video Frame Interpolation (VFI) aims at grasping the effective entities' behavior through space-time, given a sparse depiction of a scene (i.e., two frames in chronological order) [30, 64, 71, 73, 76]. Unfortunately, it suffers from the fundamentally ill-posed aspects of reconstructing continuous-time dynamics from sparse samples. The goal is to interpolate between the two instances and create a middle frame that accurately presents the state of the scene in between the two known time instants. Nonetheless, motion estimation itself is fragile due to the assumption of linear motion, which, in challenging environments, does not hold, resulting in limited temporal coherence. Lastly, most existing approaches interpolate by warping or blending frames, based on the pixel displacements provided by underlying methods.

Optical Flow (OF) measures the intensity of movement, telling us how much an object moved across space, given a certain time displacement [3, 12, 21, 35, 47]. The dense OF mask yielded by such an algorithm illustrates the behavior of each individual pixel (i.e., what trajectory it follows and with what speed it travels through space). However, OF provides us with correspondences, while VFI requires reconstruction under uncertainty. Another important aspect is that flow does not encode higher-order motion (accelerations).

In this challenge we invited the participants to come up with their alternatives to classical methods, so that they could mitigate the shortcomings of aforementioned techniques, and therefore identify the most suitable approach according to the individual challenge track requirements.

In the first track, the participants were given two sharp RGB images (i.e., the Left (L) and the Right (R)) and the only sharp ground-truth RGB image (Middle (M) - considered at the middle temporal distance between the L and R frames). Their task was to train their models to generate the

state of a single intermediary scene.

In the second track, two sharp RGB images were given (i.e., the Left (L), and the Right (R)), together with multiple sharp ground-truth RGB images in between the two (Middle Images (M1, M2, ..., M16)). The task targeted the extension of the temporal domain from discrete to contiguous and had the participants generate 16 intermediary scene states. Compared to the classic $2\times$ interpolation, this extreme $16\times$ challenge required the highest-granularity precision of the models used.

Our proposed dataset [10] contains sharp images captured with a high frame rate camera [56]. The sharp ground truth shots are taken at an exposure time of 1/1000 seconds. Therefore, the OF difference between two consecutive frames is almost insignificant in most scenarios. We select a suitable time displacement between every two input images to adopt an OF magnitude similar to the ones present in public datasets.

This challenge is one of the challenges associated with the NTIRE 2026 Workshop on: deepfake detection [20], high-resolution depth [70], multi-exposure image fusion [41], AI flash portrait [17], professional image quality assessment [39], light field super-resolution [62], 3D content super-resolution [61], bitstream-corrupted video restoration [77], X-AIGC quality assessment [33], shadow removal [59], ambient lighting normalization [58], controllable Bokeh rendering [44], rip current detection and segmentation [13], low light image enhancement [11], Night-time dehazing [1, 2], learned ISP with unpaired data [38], short-form UGC video restoration [27], rain-drop removal for dual-focused images [28], image super-resolution (x4) [9], photography retouching transfer [14], mobile real-world super-resolution [26], remote sensing infrared super-resolution [31], AI-Generated image detection [19], cross-domain few-shot object detection [40], financial receipt restoration and reasoning [16], real-world face restoration [60], reflection removal [8], anomaly detection of face enhancement [75], video saliency prediction [36], efficient super-resolution [42], 3d restoration and reconstruction in adverse conditions [32], image denoising [53], blind computational aberration correction [55], event-based image deblurring [54], efficient burst HDR and restoration [37], low-light enhancement: 'twilight cowboy' [23], and efficient low light image enhancement [68].

2. Challenge

2.1. Challenge Data

The challenge utilizes two novel datasets, MIORe and VAR-MIORe [10], both designed to capture complex motion patterns. As specified on the Codabench competition page [65], the proposed data splits contain challenging video sequences, from which we selected a snapshot

consisting in 3 (L+M+R) sharp frames for track 1, and 18 (L+M1+M2+...+M16+R) sharp frames for track 2, recorded with the high-framerate CHRONOS 2.1-HD camera [56].

MIORe emphasizes diverse and realistic motion patterns, incorporating scene motion sampled at a mild to moderate magnitude, so that it resembles the other related works. In contrast, VAR-MIORe extends this design toward extreme motion conditions, simulating scenarios with OF up to four times stronger than in MIORe. Together, the datasets provide a comprehensive testbed spanning both moderate and extreme OF regimes.

2.2. Track 1: Classic

Track 1 leverages MIORe to evaluate video frame interpolation methods under motion magnitudes comparable to existing benchmarks. The key novelty lies in the diversity of motion patterns: although the intensity varies within each scene, values are normalized via precomputed mean and maximum OF statistics. This ensures consistent difficulty while retaining scene variability. The dataset preserves the sharpness of all samples instead of blurring them, since that would give away essential descriptors of motion trajectories and intensities at the expense of losing peak signal which could be hallucinated using large visual models.

The training split contains 263 independent video sequences, each of 12 sharp images, totaling 3156 full-HD resolution (1920×1080, 8-bit RGB) samples. The validation split consists of 18 video sequences, for which only the Left (L) and Right (R) input images are given, and the testing split includes 51 such sequences. The offered training, validation, and testing sequences are mutually exclusive. Ground-truth labels for the validation and testing sequences are only available to the organizers, for scoring purposes.

2.3. Track 2: Extreme

Track 2 is built from VAR-MIORe and focuses on extreme motion scenarios largely unexplored in prior works. The captured sequences exhibit peak motion magnitudes of up to four times the severity of Track 1. This setting is designed to benchmark robustness under severe motion degradation, posing a more challenging restoration problem.

The training set comprises 263 sequences, each organized as an 18-tuple consisting of left and right input frames (L+R) along with 16 intermediate ground-truth frames (M1, ..., M16), yielding a total of 4,734 samples. The validation and test sets contain 18 and 51 sequences, respectively, where only the L+R frames are provided. All splits follow the same resolution and data format as in Track 1.

2.4. Evaluation

This challenge was a double-track competition with the same ranking criteria for both tracks. The following are the evaluation criteria:

- The reconstruction fidelity in terms of PSNR;
- The Structured Similarity Index (SSIM) [63] score;
- The LPIPS [74] distance between restorations and the ground-truth images. We used ImageNet pre-trained AlexNet [25] for LPIPS feature extraction.
- For the second track, the Mean Opinion Score (MOS) was used, since the task had a higher complexity that could not have been fully measured using classical metrics

2.5. Challenge phases

1. **Development phase:** In this phase, participants were given access to the description of the challenge, together with a set of images for Track 1 and Track 2 to train their models.
2. **Validation phase:** During this phase, to validate the solutions, participants received a set of input images for Track 1 and Track 2 from the validation split, with the ground truths not shared. To enable measurement of validation performance, a Codabench server [65] was set up, comparing submitted images uploaded by each participant with private reference images.
3. **Final phase:** Sets of input images for Track 1 and Track 2 from the testing split were sent to challenge participants, with evaluation performed by the Codabench server. The fine-tuning of the test set was restricted by limiting user submissions. Finally, participants received a submission template, with instructions for the final preparation of the submission. Each team provided their method description, corresponding code, information regarding team members, affiliations, and the final set of restored images, corresponding to the input from the testing split.

3. Challenge Results

We received 8 valid submissions for track 1 and 4 valid submissions for track 2 that are presented in Section 4.

Tables 1 and 2 provide the quantitative evaluation of the submitted results for both tracks, respectively. Particular rankings along each metric evaluated are provided as subscripts. The solutions achieve a significant performance level, in terms of reconstruction fidelity and perceptual quality, with the provided evaluations being backed by the results provided for visual comparison (see Figures 1 and 2 for the two tracks, respectively).

4. Challenge Methods

4.1. Hawkeye for Track 1 and Track 2

Method. Team Hawkeye participated in both challenge tracks with two different solutions. For Track 1, the team [48] proposes a motion-guided multi-scale fusion model built on EMA-VFI [72] (Figure 3a). The input pair is processed at three scales, and the resulting predictions are fused

Table 1. Evaluation and Rankings in the NTIRE 2026 High FPS Video Frame Interpolation Challenge - Track 1. The superior indices next to the metric scores indicate the respective standings of teams based on their performances in that corresponding metric on the challenge’s test dataset. The “Final Rank” is computed based on the PSNR performance.

Team	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Params. (M)	Runtime (s)	Device	Rank
hawkeye	26.34 ¹	0.7356 ³	0.4064 ⁸	73.89	2.05	RTX 3090	1
KLETech-CEVI	26.32 ²	0.7427 ¹	0.4008 ⁶	66	4.9	RTX 6000	2
KLETech-CEVI_XR	26.31 ³	0.7425 ²	0.4017 ⁷	63	5.1	A6000	3
micamera	26.08 ⁴	0.7241 ⁴	0.2944 ²	30	-	V100	4
Tencent Video	25.38 ⁵	0.7014 ⁸	0.1684 ¹	19	0.53	4 $\times$ A100	5
BuptMM	25.09 ⁶	0.7147 ⁷	0.3162 ³	30	0.5	RTX 4090	6
Eidolon	25.08 ⁷	0.7198 ⁵	0.3436 ⁴	40	0.52	A6000	7
NTR	24.88 ⁸	0.7166 ⁶	0.3519 ⁵	65.66	1.07	A10	8

Table 2. Evaluation and Rankings in the NTIRE 2026 High FPS Video Frame Interpolation Challenge - Track 2 Extreme. The superior indices next to the evaluation score values indicate the teams’ ranking for that particular metric, based on their performances on the challenge’s test dataset. The “Final Rank” represents a metric based on the Mean Opinion Score (MOS), performed through a user study.

Team	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	MOS $\uparrow$	Params. (M)	Runtime (s)	Device	Perceptual Rk.
hawkeye	23.16 ³	0.6418 ³	0.3774 ³	6.58	26.63	24	RTX 3090	1
Tencent Video	24.59 ¹	0.6727 ²	0.1663 ¹	5.08	19	3.7	4 $\times$ A100	2
NTR	24.40 ²	0.6993 ¹	0.4532 ⁴	3.25	65.66	8	A6000	3
Eidolon	22.63 ⁴	0.6366 ⁴	0.2626 ²	1.67	9.8	1.8	A6000	4

by a lightweight U-Net conditioned on flow-derived motion maps, which improves robustness to large displacement while preserving fine details. The model is trained on the official NTIRE challenge data and further strengthened with the X4K1000FPS dataset introduced by XVFI [49].

For Track 2, the team uses BiMotionHD (Figure 3b), a perceptual pipeline designed for 16 $\times$ 1080p interpolation under MOS evaluation. It performs bidirectional diffusion-based interpolation with a Stable Video Diffusion backbone [5] following ViBiDSampler [69], restores full-resolution outputs with a Neural Heat Fields super-resolution module based on Thera [4], and applies a Codex CLI based candidate-selection workflow to choose the most perceptually favorable outputs for each sample. This decoupled design targets temporally coherent motion synthesis at reduced resolution and aliasing-free recovery at 1080p.

Training Details. Track 1 is trained in two stages: the EMA-VFI backbone is first frozen while only the fusion module is optimized, and the full network is then jointly fine-tuned on the NTIRE training data together with X4K1000FPS. Track 2 keeps the diffusion interpolation stage training-free and fine-tunes only the post-processing super-resolution module on the official challenge data, without using extra data. Overall, the two submissions reflect different optimization targets: a distortion-oriented multi-scale regression model for Track 1 and a perception-

oriented generative pipeline for Track 2.

4.2. KLETech-CEVI for Track 1

Method. The authors proposed a refined **VFIMamba-Large** framework for high-FPS video frame interpolation, as depicted in Figure 4. The methodology is structured into three primary stages:

1. Hybrid CNN-SSM Feature Extraction: A hierarchical encoder (MambaFeature) processes input frames $\mathbf{I}_0$ and $\mathbf{I}_1$. It utilizes convolutional stages for spatial features and **BiMambaBlocks** (incorporating S6 scans) for long-range temporal modeling. This mixed architecture achieves $\mathcal{O}(HW)$ complexity, making it suitable for FHD resolution.

2. Flow Estimation and Frame Synthesis: The system performs coarse-to-fine bidirectional flow estimation. The authors corrected the forward pass by re-enabling **IFBlock modules** for local motion refinement: $(\mathbf{F}, \mathbf{M}, \{M_i\}_{i=1}^2, \hat{I}_t) = \mathcal{N}(\mathbf{I}_0 \oplus \mathbf{I}_1; \tau = 0.5, \lambda = 1.0, \text{local} = 2)$. The final frame is synthesized via a 4-level U-Net (RefineNet) that predicts a residual r to produce the final output $\hat{I}_t = \text{clamp}(M_{\text{final}} + r, 0, 1)$.

3. Training Rectifications: The team introduced multi-scale supervision to ensure train-inference consistency. They replaced the standard sharpness loss with a **Laplacian Pyramid Loss** ($\mathcal{L}_{\text{Lap}}$) using Charbonnier distance across five levels: $\mathcal{L} = \mathcal{L}_{\text{Lap}}(\hat{I}_t, I^*) + 0.5 \sum_{i=1}^2 \mathcal{L}_{\text{Lap}}(M_i, I^*)$. This objective enforces structural accuracy at multiple reso-



Figure 1. Visual comparison of top-performing methods on representative samples from the NTIRE 2026 High FPS Video Frame Interpolation Challenge (Track 1, testing split). Given the input frame pairs (L+R), competing approaches exhibit notable differences in reconstructing intermediate content under challenging conditions such as large motion, occlusion, and fine-structure preservation. Leading methods generally achieve sharper spatial details and improved edge fidelity, while effectively mitigating common artifacts including ghosting, blurring, and temporal inconsistency. Nevertheless, subtle variations persist across methods, particularly in regions with complex motion or thin structures.

lutions, significantly improving the reconstruction of high-frequency details.

Training Details. Initialized from VFIMamba-Large Step2 [67]; fine-tuned on 3,419 mixed-step C18 triplets (both $\Delta t=1$ step-1 and $\Delta t=2$ step-2 strides, mixed to balance precise pixel accuracy and large-motion robustness) using AdamW [34] ($\beta_1=0.9$, $\beta_2=0.999$, wd 10^{-4}), cosine LR, and gradient clipping (max norm 1.0).

4.3. KLETech-CEVLR for Track 1

Method The authors employ the **VFIMamba-Large** architecture (Figure 5), which utilizes a three-stage pipeline for high-fidelity video frame interpolation:

- **Feature Extraction:** A five-stage hierarchical encoder combines Conv blocks (Stages 1–3) for spatial features with **Mixed-SSM Blocks** (Stages 4–5) for long-range temporal modeling using multi-directional S6 scanning.
- **Flow Estimation:** Bidirectional flow and occlusion masks are predicted coarse-to-fine via Head modules and **IFBlock** refinement, producing intermediate blended outputs $\{M_i\}$.
- **Frame Synthesis:** A 4-level U-Net aggregates warped frames and multi-scale features to predict a residual, refining the final mask-blended output.

Multi-Objective Training: The core contribution is a composite loss function ($\mathcal{L}$) that targets specific competition metrics through multi-scale supervision: $\mathcal{L} = \mathcal{L}_{\text{pixel}} + 0.02\mathcal{L}_{\text{VGG}} + 0.01\mathcal{L}_{\text{Cen}}$. The components include:

1. **Laplacian Pyramid Loss ($\mathcal{L}_{\text{Lap}}$):** A 5-level pyramid loss using Charbonnier distance to maximize pixel fidelity and PSNR.
2. **VGG Perceptual Loss ($\mathcal{L}_{\text{VGG}}$):** Minimizes L1 distance in VGG-19 feature space (relu1_2, 2_2, 3_3) to improve LPIPS scores.
3. **Census Loss ($\mathcal{L}_{\text{Cen}}$):** Penalizes soft Hamming distance between illumination-invariant census signatures to enhance structural similarity (SSIM).

Training Details The team finetuned their model starting from the VFIMamba-Large Step2 checkpoint [66] and used the methodology: AdamW [34] ($\beta_1=0.9$, $\beta_2=0.999$, wd 10^{-4}); Cosine LR annealing: $5 \times 10^{-6} \rightarrow 5 \times 10^{-7}$; 40 epochs, crop 256×256 , batch size 8; Mixed-step triplets ($\Delta t=1$ and $\Delta t=2$) from the C18 training split; Augmentation: H/V flip (50%), temporal reversal (50%), 90° rotation (30%); Gradient clipping: max norm 1.0.

4.4. micamera for Track 1

Method The team adopts a two-branch, flow-based VFI framework that integrates GIMM-VFI [18] and BiM-

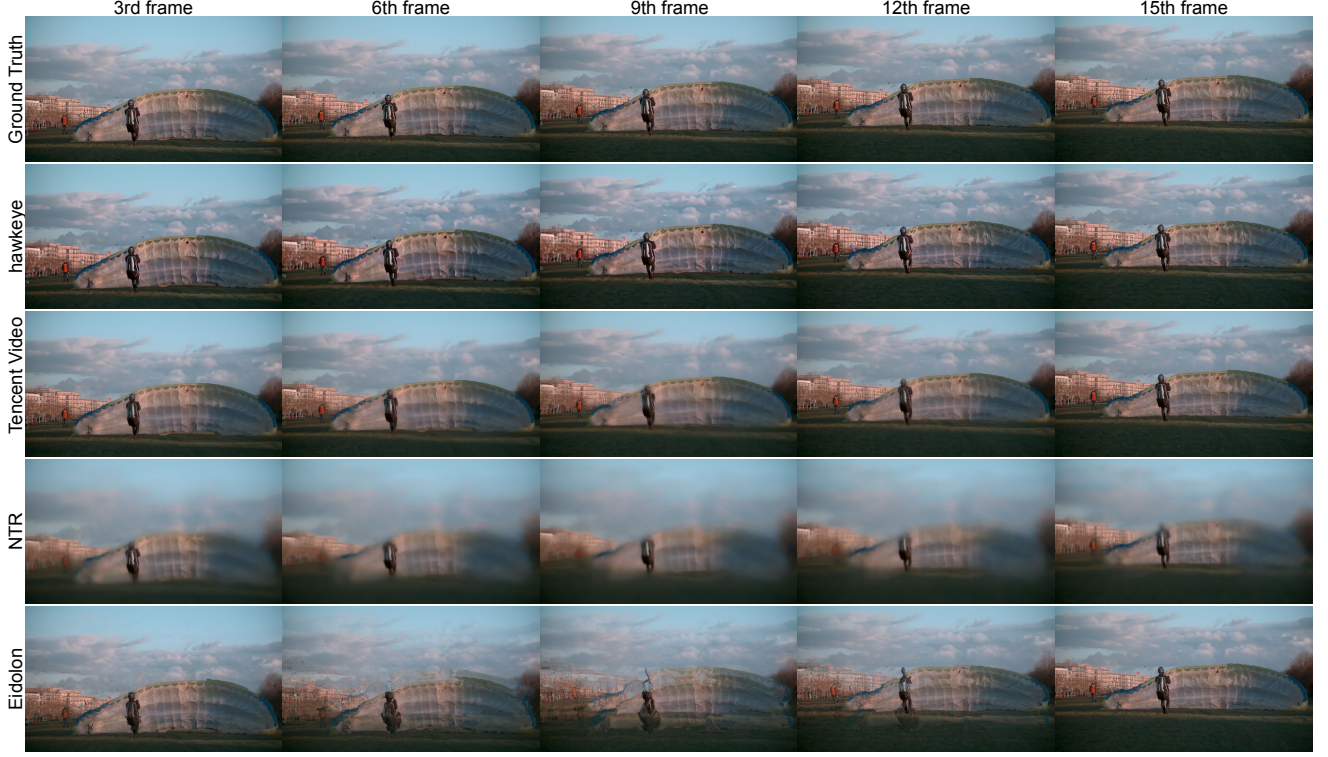


Figure 2. Visual comparison of top-performing methods on representative samples from the NTIRE 2026 High FPS Video Frame Interpolation Challenge (Track 2, testing split). This track involves extreme interpolation ($\times 16$), requiring synthesis of multiple intermediate frames across large temporal gaps. As shown, all methods operate on the same input frames but exhibit varying capabilities in modeling long-range motion and maintaining temporal coherence. Top solutions generally produce visually sharp and temporally consistent results across intermediate timesteps (e.g., 3rd, 6th, 9th, 12th, and 15th frames), demonstrating strong generalization to large displacements and complex scene dynamics. However, differences become more pronounced at farther timesteps, where errors accumulate. Some methods exhibit motion drift, structural distortion, or temporal flickering, particularly in regions with fast motion, occlusions, or fine textures. In contrast, more robust approaches better preserve object boundaries and motion trajectories while reducing ghosting artifacts. These results highlight the increasing difficulty of high-FPS interpolation under large temporal intervals and the importance of effective temporal modeling for maintaining both spatial fidelity and consistency across frames.

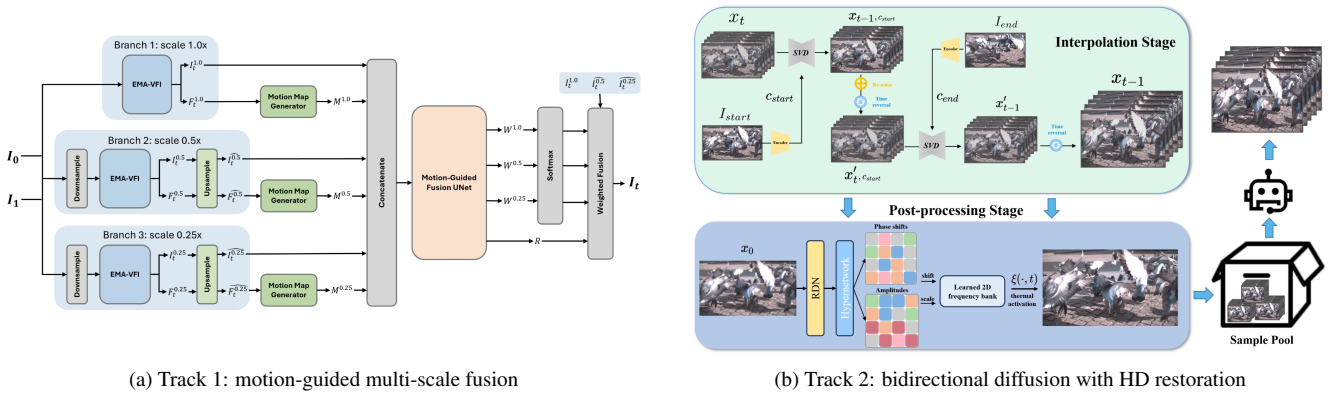


Figure 3. Overview of the two Hawkeye submissions included in the NTIRE 2026 VFI challenge report

VFI [45]. The GIMM-VFI branch employs an implicit neural representation to model motion continuously and estimate timestamp-conditioned bidirectional flows ($F_{t \rightarrow 0}$,

$F_{t \rightarrow 1}$), providing robustness to large motions; it is executed at $0.5\times$ resolution to mitigate artifacts. The BiM-VFI branch learns VFI-oriented bidirectional flows with teacher-

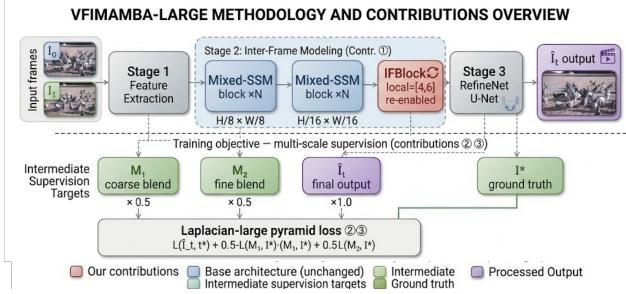


Figure 4. VFIMamba-Large pipeline with Team KLETech-CEVI contributions. **Top:** Inference pipeline — Stage 1 (feature extraction), Stage 2 (inter-frame modeling with re-enabled IFBlock ①), Stage 3 (RefineNet synthesis). **Bottom:** Training objective showing multi-scale supervision ② applying LapLoss ③ to M_1 ($\times 0.5$), M_2 ($\times 0.5$), and $\hat{I}_t$ ($\times 1.0$) against ground truth I^* .

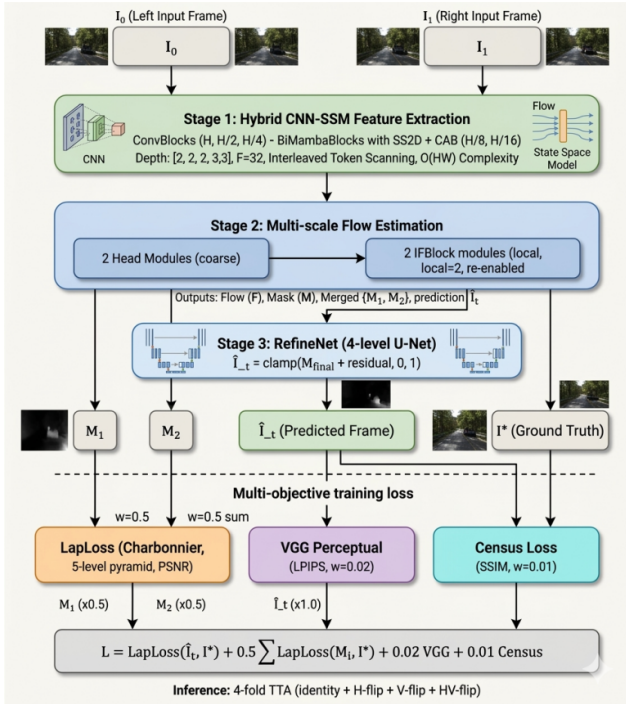


Figure 5. Methodology proposed by Team KLETech-CEVI based on VFIMamba for High FPS VFI.

flow supervision, enabling full-resolution inference with sharper details but increased sensitivity to occlusions. The two predictions are fused via α -blending, balancing high-frequency fidelity and ghosting artifacts.

Training Details Training uses only the official dataset. Pseudo optical flows are generated using a pre-trained Flowformer model to supervise the INR module, followed by joint fine-tuning with triplet sampling over sliding windows. Loss functions (L1/L2 and LPIPS) and blending

weights are selected based on validation performance. Ablations include blending ratios, GIMM-VFI inference resolution, and patch-based inference, the latter being less effective under strong global motion.

4.5. Tencent Video for Track 1 and Track 2

Method The team builds upon EMA-VFI [72], a hybrid CNN-Transformer architecture for video frame interpolation. EMA-VFI employs a MotionFormer backbone consisting of three convolutional stages and two Transformer stages, each equipped with shifted window attention and inter-frame cross-attention for jointly extracting multi-scale motion and appearance features. A coarse-to-fine flow estimation module subsequently predicts bidirectional optical flow fields and blending masks from the extracted features.

Two key modifications are introduced to the original pipeline. (1) Adaptive-Resolution Optical Flow Estimation. Input frames are downsampled before feature extraction and flow prediction, where the downsampling factor is adaptively determined based on the estimated motion magnitude—larger motions lead to stronger downsampling. This effectively enlarges the receptive field for handling large displacements while reducing computation. The estimated flow is then upsampled back to perform warping and synthesis at full resolution, preserving spatial fidelity. (2) Removal of RefineNet. The post-synthesis refinement network is removed. While RefineNet can improve per-frame fidelity, it introduces sporadic artifact frames under complex motion. By directly using the mask-weighted blending output $\hat{I}_t = M \odot \hat{I}_{0 \rightarrow t} + (1 - M) \odot \hat{I}_{1 \rightarrow t}$ as the final prediction, significantly better temporal stability is achieved. The proposed pipeline is depicted in Figure 6.

Training Details The model is trained on $4 \times$ A100 GPUs in two stages using AdamW (learning rate 2×10^{-4} , weight decay 10^{-4}) with cosine annealing warm restarts. In Stage 1 (pre-train, 150 epochs), the model is trained from scratch on a mixture of Vimeo-90K Septuplet [67], LAVIB [51], a self-collected film/TV/documentary dataset, and a synthetic motion dataset with random background and foreground layers moving along predefined trajectories, using L1 loss only. In Stage 2 (fine-tune, 100 epochs), the model is fine-tuned on the competition training set with a composite loss: $0.5 \mathcal{L}_{L1} + 0.5 \mathcal{L}_{\text{MS-SSIM}} + 0.5 \mathcal{L}_{\text{Laplacian}} + \mathcal{L}_{\text{LPIPS}}$. Data augmentation in both stages includes scale augmentation, simulated camera motion, geometric overlay, subtitle overlay, color adjustment, and random flipping/rotation. At inference, input frames are downsampled and padded to a multiple of 32; flow estimation is performed at reduced resolution, then upsampled for full-resolution warping and synthesis.

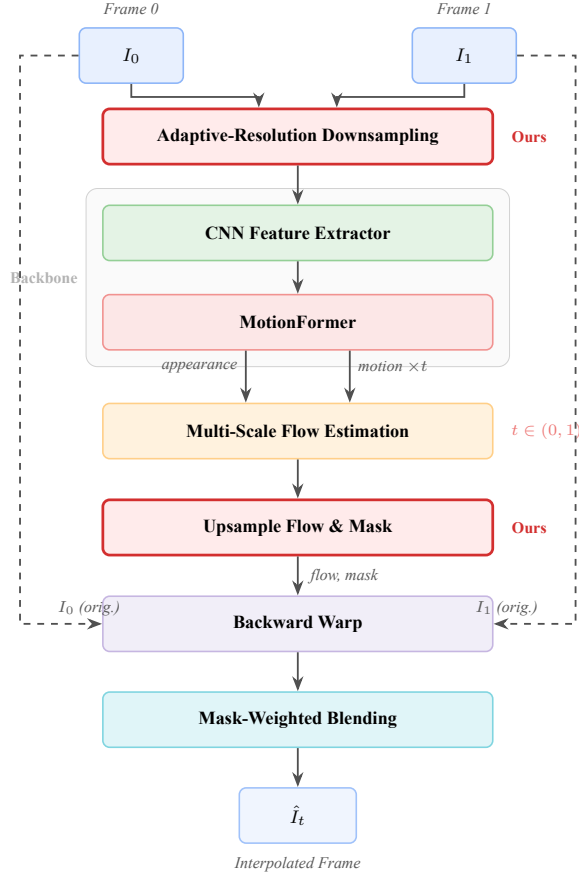


Figure 6. Proposed method of Tencent Video: Red-highlighted modules indicate our modifications over EMA-VFI: adaptive-resolution downsampling and refinement-free synthesis.

4.6. BuptMM for Track 1

Method The team adopts AMT (All-Pairs Multi-Field Transforms) [29] as the backbone. Given input frames I_0 and I_1 , AMT constructs all-pairs correlation volumes and applies multi-field transforms to jointly estimate bidirectional flows and synthesize the intermediate frame $\hat{I}_{0.5}$ in a single stage. Training uses a combination of Charbonnier and Ternary losses. The model is initialized from AMT-G pre-trained on Vimeo90K and fine-tuned on the NTIRE 2026 training set only. Inference is performed at full resolution with padding to multiples of 16, and midpoint interpolation ($t = 0.5$).

Training Details Training is conducted for 300 epochs on 4×RTX 4090 with a batch size of 128. Triplets (I_i, I_{i+1}, I_{i+2}) are sampled with stride 1, and a 9:1 split by video ID is used for training and validation. Optimization uses AdamW with cosine annealing ($2 \times 10^{-5} \rightarrow 1 \times 10^{-6}$).

Random 224×224 crops and standard augmentations are applied. Gradient clipping ensures stability. Checkpoints are selected based on validation PSNR and the final epoch.

4.7. Eidolon for Track 1 and Track 2

Method For Track 1, the team builds upon EMA-VFI [72] and proposes OptEMA-VFI, retaining the original architecture while improving the optimization objective. A composite loss integrates Charbonnier, SSIM, Laplacian pyramid, and gradient terms to enhance fidelity, perceptual quality, and edge consistency, along with multi-scale optical flow smoothness regularization for temporal coherence. Intermediate predictions are jointly supervised under the same objective.

For Track 2, the team extends SuperSloMo [22] with a timestep-embedding mechanism (TTE-SuperSloMo) to better encode temporal positions. This enables improved modeling of long-range dependencies compared to linear scaling. Training employs a combination of L_1 , perceptual (VGG16), warping consistency, and flow smoothness losses.

Training Details For Track 1, training uses the official dataset with spatial and photometric augmentations, including color jitter, blur, and noise. A 3×3 grid-based cropping strategy with 256×256 patches is adopted, and training runs for 300 epochs on a single RTX A6000.

For Track 2, triplets (I_0, I_t, I_1) are constructed with standard preprocessing. The model is trained for 200 epochs on a single GPU, with a total runtime of approximately 1.5–2 hours.

4.8. NTR: Track 1 and Track 2

Method The team is based on EMA-VFI [72], a hybrid CNN–Transformer framework for VFI, applied to both Track 1 ($t=0.5$) and Track 2 ($t_k=k/17$). The backbone (MotionFormer, 17.8M) is a hierarchical Transformer with inter-frame attention for joint motion–appearance encoding. A coarse-to-fine MultiScaleFlow decoder estimates bidirectional flows and blending masks, followed by a U-Net refinement module.

Track 1 adopts the fixed-timestep variant, while Track 2 uses a time-aware variant conditioned on t_k . For efficiency, Track 2 reuses shared features across all timesteps, reducing the computation by $\sim 8\times$. Test-time augmentation [57] averages the predictions from the original and 180° -rotated inputs.

Training Details Models are initialized from EMA-VFI pretrained on Vimeo-90K and fine-tuned with AdamW (cosine decay, 300 epochs). Training uses a 5-level Laplacian pyramid loss applied to both final and intermediate predic-

tions. Standard augmentations (random crops, flips, rotations, channel and temporal reversal) are applied.

Track 1 is trained on 510 triplets with batch size 18. Track 2 is trained on combined data from both tracks with multi-stride augmentation ($\sim 17.8k$ triplets) and batch size 8.

5. Discussion and Conclusion

The video frame interpolation (VFI) task on a subset of our MIORe dataset [10] has presented a real challenge to participants, through its array of difficult motion types, the composition of scenes, and the range of motion magnitude. The first track was more manageable and was solved close to reality by teams such as *micamera* and *Tencent Video*. Although their perceptual quality is better, the conclusion of this track further supports the perception-distortion trade-off claim [6]. In the future, different evaluation means will be used to ensure a better alignment of quantitative and qualitative metrics. The second track could not have been fully solved correctly due to its step-by-step generation requirement. We have observed that the employed VFI methods that either warped or blended the scene did well in the first track, but severely underperformed here. Generative methods took the lead and outperformed the other approaches by a significant margin. Nevertheless, it did not lack errors, since the motion in its entirety could not have been fully reproduced correctly. In future challenge editions, additional guidance cues may be required to instruct the models to follow a certain pattern. As a result, the High-FPS Video Frame Interpolation NTIRE 2026 challenge presented several novel and creative ideas to advance the state of knowledge, and attempted to raise the awareness of the importance of motion decomposition into its elements to correctly estimate the optical flow and restore intermediary video frames.

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